# GPT-6.1 Sol

# 10MAA End of Year Revision Examination 2026

# Marking Schedule

**General comments**

Accept equivalent methods and exact answers unless a particular form or accuracy is requested. Require working for method marks. Correct algebraic expressions may themselves show the method. Allow follow-through where a subsequent method is valid. Do not penalise the same arithmetic error twice. Penalise incorrect rounding only once across the paper. Retain unrounded values in calculations.

## Algebraic Expressions & Equations • 31 marks

| Question | Working and answer | Marks | Comments |
|---|---|---|---|
| 1.(a) | $(x-4)^2 = (x-4)(x-4)$<br>$= x^2 - 8x + 16$ | 2 | 1 for two correct cross-products or equivalent expansion.<br>1 for final expression. |
| (b) | $\dfrac{x^2-16}{x-4} = \dfrac{(x-4)(x+4)}{x-4} = x+4$ | 2 | 1 for difference-of-squares factorisation.<br>1 for cancellation and $x+4$. |
| (c) | $x^2 - 8x + 16 = x^2 + 16$<br>$-8x = 0$, so $\boldsymbol{x = 0}$. | 2 | 1 for equating the correct expansion to the student's expression.<br>1 for 0. |
| 2.(a) | $(16m^{12})^{\frac{1}{4}} = 2m^3$ | 2 | 1 for fourth root of 16.<br>1 for $m^3$. |
| (b) | $3^{2a} \times 3^{3b} = 3^{22}$<br>$2a + 3b = 22$, $a = b + 1$<br>$2(b+1) + 3b = 22$<br>$\boldsymbol{b = 4}$, $\boldsymbol{a = 5}$. | 3 | 1 for a common base.<br>1 for the equation in one variable.<br>1 for both values. |
| 3.(a) | First differences: $7, 11, 15$.<br>Second differences: $4, 4$, which are constant. | 2 | 1 for all first differences.<br>1 for second differences and conclusion. |
| (b) | $T = an^2 + bn + c$<br>$2a = 4$, so $a = 2$.<br>$2 + b + c = 7$, $8 + 2b + c = 14$<br>$b = 1$, $c = 4$<br>$\boldsymbol{T = 2n^2 + n + 4}$. | 3 | 1 for quadratic coefficient 2.<br>1 for finding the remaining coefficients.<br>1 for the complete formula. Accept other valid methods. |
| (c) | $T(10) = 2(100) + 10 + 4 =$ **214 ms**. | 1 | Allow follow-through from (b). |
| 4.(a) | $p + r = 20$<br>$3p + 5r = 78$ | 2 | 1 for each equation. |
| (b) | $3p + 3r = 60$<br>$2r = 18$, so $r = 9$ and $p = 11$.<br>**11 Python stickers and 9 Rust stickers**. | 2 | 1 for valid elimination or substitution.<br>1 for both counts, correctly identified. |

| Algebraic Expressions & Equations continued | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 5. | $2(3x-5) > 5x + k$ gives $x > k + 10$.<br>True at $x = 4$: $4 > k + 10$, so $k \leftarrow 6$.<br>False at $x = 1$: $1 \leq k + 10$, so $k \geq -9$.<br>Thus $-9 \leq k \leftarrow 6$.<br>$\boldsymbol{k = -9, -8, -7}$. | 4 | 1 for correct simplification or substitution at both test values.<br>1 for upper bound $k \leftarrow 6$.<br>1 for lower bound $k \geq -9$.<br>1 for the complete integer list. |
| 6.(a) | The enlarged dimensions are $(n + 2)$ and $(n + 9)$.<br>$\boldsymbol{(n+2)(n+9) = 198}$. | 2 | 1 for both enlarged dimensions.<br>1 for the area equation. |
| (b) | $n^2 + 11n - 180 = 0$<br>$(n + 20)(n - 9) = 0$<br>$n = -20$ or $n = 9$.<br>Reject $-20$ because a cell count cannot be negative.<br>Original dimensions: **9 cells by 16 cells**. | 4 | 1 for the correct quadratic.<br>1 for solving it.<br>1 for selecting 9 with a contextual reason.<br>1 for both original dimensions. |

| Statistics & Probability • 27 marks | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 1.(a) | $36 - 18 =$ **18 seconds**. | 1 | Correct difference. |
| (b) | Linux: $22 - 16 = 6$ s.<br>macOS: $30 - 20 = 10$ s.<br>**Linux has the smaller interquartile range**. | 2 | 1 for both interquartile ranges.<br>1 for comparison identifying Linux. |
| (c) | macOS: **approximately 75%**.<br>Linux: **100%**. | 2 | 1 for each. Accept a quartile-based estimate for macOS. |
| (d) | **No**. The slowest Linux run is 28 s, while the fastest Windows run is 20 s. | 1 | A valid comparison or counterexample is required. Merely stating that the median is lower does not establish the claim. |
| 2.(a) | $2 + 5 + 7 + 9 + k + 15 = 6(8)$<br>$38 + k = 48$, so $\boldsymbol{k = 10}$. | 2 | 1 for using total $= 6 \times \text{mean}$.<br>1 for 10. |
| (b) | Median before: $\dfrac{7 + 9}{2} = 8$.<br>Median after: **8**.<br>Changing the largest value does not change the third and fourth values. | 2 | 1 for both medians.<br>1 for explaining that the middle two values are unchanged. |

| Statistics & Probability continued | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 3.(a) | First card: Rust $\frac{3}{9}$; not Rust $\frac{6}{9}$.<br>After Rust: Rust $\frac{2}{8}$; not Rust $\frac{6}{8}$.<br>After not Rust: Rust $\frac{3}{8}$; not Rust $\frac{5}{8}$. | 3 | 1 for each complete pair of branches on a correctly structured two-stage tree. Accept simplified fractions. |
| (b) | $P(\text{same}) = \frac{4}{9}\times\frac{3}{8} + \frac{3}{9}\times\frac{2}{8} + \frac{2}{9}\times\frac{1}{8}$<br>$= \frac{12+6+2}{72} = \frac{20}{72} = \mathbf{\frac{5}{18}}$. | 3 | 1 for correct Python and Rust paths.<br>1 for the C path and addition of all three.<br>1 for $\frac{5}{18}$. Accept a combinations method. |
| (c) | $P(\text{both Rust}) = \frac{3}{9}\times\frac{2}{8} = \frac{1}{12}$<br>$P(\text{both Rust} \mid \text{same}) = \frac{\frac{1}{12}}{\frac{5}{18}} = \mathbf{\frac{3}{10}}$. | 2 | 1 for using the same-language event as the condition.<br>1 for $\frac{3}{10}$. |
| (d) | $180 \times \frac{5}{18} =$ **50 pairs**. | 1 | Allow follow-through from (b). |
| 4.(a) | **Diagram B**.<br>Its bar areas are $20(0.6) = 12$, $20(0.9) = 18$, $40(0.6) = 24$, $60(0.3) = 18$, so area represents frequency.<br>In A, unequal class widths make the areas inconsistent with the frequencies. | 2 | 1 for Diagram B.<br>1 for a correct area–frequency justification. Merely naming frequency density without explaining areas earns only the first mark. |
| (b) | $20(0.6) =$ **12 projects**.<br>Assume build times are evenly spread within $40 < t \le 80$. | 2 | 1 for 12.<br>1 for the uniform-within-class assumption. |
| (c) | All 18 projects in $80 < t \le 140$ exceed 50 s. Between 0 and 24 of the projects in $40 < t \le 80$ could exceed 50 s.<br>**Least 18; greatest 42**. | 2 | 1 for each bound. No within-class assumption is allowed here. |
| (d) | Let $u = \frac{t}{2}$.<br>New class: $\mathbf{20 < u \le 40}$ **seconds**.<br>Frequency stays 24, while width becomes 20.<br>New density: $\mathbf{\frac{24}{20} = 1.2}$. | 2 | 1 for the halved interval.<br>1 for density 1.2, not 0.3. Accept the original variable name if the meaning is clear. |

## Trigonometry • 13 marks

| Question | Working and answer | Marks | Comments |
|---|---|---|---|
| 1.(a) | $\dfrac{\sin C}{14} = \dfrac{\sin 35°}{10}$<br>$\sin C = 1.4 \sin 35° = 0.803007...$<br>$C = 53.4182...°$ or $180° - 53.4182...°$<br>**53.4° or 126.6°**. | 3 | 1 for correct sine-rule equation.<br>1 for the acute value.<br>1 for the obtuse value. |
| (b) | $A = 180° - 35° - 126.5818...°$<br>$= 18.4182...°$<br>$\dfrac{BC}{\sin A} = \dfrac{10}{\sin 35°}$<br>$BC = 5.50843... =$ **5.51 cm**. | 2 | 1 for the correct remaining angle and sine-rule setup.<br>1 for BC. Allow a valid alternative method. |
| 2.(a) | $\boldsymbol{h = x \tan 55°}$. | 1 | Correct expression. |
| (b) | $\boldsymbol{x \tan 55° = (x + 20) \tan 35°}$. | 1 | An equation in $x$ only is required. |
| (c) | $x(\tan 55° - \tan 35°) = 20 \tan 35°$<br>$x = 19.2380...$<br>$h = x \tan 55° = 27.4748...$<br>**27.5 m**. | 3 | 1 for correctly isolating $x$.<br>1 for calculating $h$ from $x$.<br>1 for final height. Allow follow-through. |
| 3. | Bearing of A from B is 220°.<br>$\angle ABC = 220° - 100° = 120°$.<br>$AC^2 = 8^2 + 6^2 - 2(8)(6)\cos 120°$<br>$AC = \sqrt{148} =$ **12.2 km**. | 3 | 1 for included angle 120°.<br>1 for cosine rule.<br>1 for distance. |

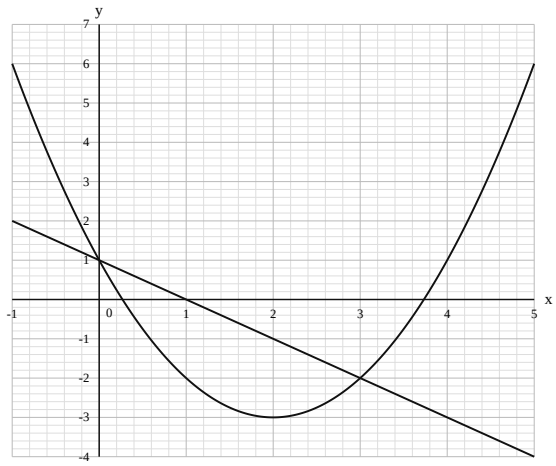
## Measurement • 16 marks

| Question | Working and answer | Marks | Comments |
|---|---|---|---|
| 1.(a) | Internal dimensions: $12 - 0.8 = 11.2$ cm,<br>$8 - 0.8 = 7.2$ cm, $5 - 0.4 = 4.6$ cm.<br>$V = 11.2(7.2)(4.6) = 370.944\text{cm}^3$<br>**371 cm³**. | 3 | 1 for reducing both horizontal dimensions by twice the wall thickness.<br>1 for internal height 4.6.<br>1 for capacity. |
| (b) | $V_{\text{external}} = 12(8)(5) = 480\text{cm}^3$<br>$V_{\text{material}} = 480 - 370.944 = 109.056\text{cm}^3$<br>**109 cm³**. | 2 | 1 for external volume minus internal capacity.<br>1 for material volume. |
| (c) | $\text{Mass} = 109.056(1.24) = 135.22944$<br>**135 g**. | 1 | Allow follow-through from (b). |

| Measurement continued | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 2.(a) | Volume scale factor $= \dfrac{729}{216} = \dfrac{27}{8}$<br>Length scale factor $= \sqrt[3]{\dfrac{27}{8}} = \dfrac{3}{2}$<br>Height $= 8 \times \dfrac{3}{2} =$ **12 cm**. | 2 | 1 for the cube root giving length scale factor 1.5.<br>1 for 12 cm. |
| (b) | Area scale factor $= \left(\dfrac{3}{2}\right)^2 = \dfrac{9}{4}$<br>Area $= 96 \times \dfrac{9}{4} =$ **216 cm²**. | 2 | 1 for squaring the length scale factor.<br>1 for area. |
| 3.(a) | $\mathbf{2.95 \le r < 3.05}$ **cm**.<br>$\mathbf{11.5 \le h < 12.5}$ **cm**. | 2 | 1 for each complete pair of bounds. |
| (b) | $V_{\text{minimum}} = \pi(2.95)^2(11.5)$<br>$= 314.406...\text{cm}^3 < 350\text{cm}^3$.<br>**The claim is not guaranteed**. | 2 | 1 for a valid lower-bound capacity calculation.<br>1 for comparison with 350 and conclusion. An attainable counterexample is also valid. |
| (c) | $V_{\text{water}} = \pi(3)^2(12) - \dfrac{1}{3}\pi(3)^2(12)$<br>$= 108\pi - 36\pi = 72\pi\text{cm}^3$<br>**226 cm³**. | 2 | 1 for cylinder volume minus cone volume.<br>1 for $72\pi$ or 226 cm³. |

| Coordinate Geometry • 12 marks | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 1.(a) | $m_{AB} = \dfrac{5-1}{4-(-4)} = \dfrac{1}{2}$<br>$m_{BC} = \dfrac{1-5}{6-4} = -2$<br>$m_{AB}m_{BC} = -1$, so $AB \perp BC$. | 3 | 1 for each gradient.<br>1 for product $-1$ and perpendicular conclusion. |
| (b) | AC is horizontal, with length $6-(-4) = 10$.<br>The perpendicular height from B is $5-1 = 4$.<br>$A = \dfrac{1}{2}(10)(4) =$ **20 square units**. | 2 | 1 for a valid base and height, or both perpendicular side lengths.<br>1 for area. |
| (c) | Reflection in $y = 1$ gives $\mathbf{(4, -3)}$. | 1 | Correct coordinates. |
| 2.(a) | $2x + 1 = -\dfrac{1}{2}x + 6$<br>$\dfrac{5}{2}x = 5$, so $x = 2$, $y = 5$.<br>**P** $\mathbf{(2, 5)}$. | 2 | 1 for solving the simultaneous equations.<br>1 for both coordinates. |
| (b) | At A: $-3 = 2x + 1$, so $x = -2$.<br>At B: $-3 = -\dfrac{1}{2}x + 6$, so $x = 18$.<br>$AB = 18 - (-2) =$ **20 units**. | 2 | 1 for both x-coordinates.<br>1 for distance. |
| (c) | $AK = 2-(-2) = 4$, $KB = 18-2 = 16$.<br>The triangles have equal height PK.<br>Area ratio $= 4 : 16 =$ $\mathbf{1 : 4}$. | 2 | 1 for correct bases and common-height reasoning, or correct individual areas 16 and 64.<br>1 for ratio 1 : 4. |

| Graphs of Quadratic Functions • 23 marks | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 1.(a) | $y = a(x-2)^2 + 9$<br>At $(0, 5)$: $5 = 4a + 9$, so $a = -1$.<br>$\boldsymbol{y = -(x-2)^2 + 9}$. | 3 | 1 for the correct vertex form.<br>1 for substituting $(0, 5)$ and finding $a$.<br>1 for the equation. |
| (b) | $0 = -(x-2)^2 + 9$<br>$(x-2)^2 = 9$, so $x = 2 \pm 3$.<br>$\boldsymbol{x = -1}$ **m and** $\boldsymbol{x = 5}$ **m**.<br>Both are valid positions: 1 m west and 5 m east of the post. | 3 | 1 for setting $y = 0$ and rearranging.<br>1 for both roots.<br>1 for retaining and interpreting both positions. Accept $(-1, 0)$ and $(5, 0)$. |
| (c) | $-(x-2)^2 + 9 \geq 8$<br>$(x-2)^2 \leq 1$<br>$\boldsymbol{1 \leq x \leq 3}$. | 3 | 1 for the height inequality.<br>1 for boundary values 1 and 3.<br>1 for the correct closed interval between them. |
| 2.(a) | $2(n^2 + 3n) + 12$<br>$= 2\left(\left(n + \frac{3}{2}\right)^2 - \frac{9}{4}\right) + 12$<br>$\boldsymbol{2\left(n + \frac{3}{2}\right)^2 + \frac{15}{2}}$. | 2 | 1 for the correct square term.<br>1 for constant $\frac{15}{2}$. Accept decimal equivalents. |
| (b) | $2n^2 + 6n + 12 \leq 200$<br>Boundary: $n^2 + 3n - 94 = 0$.<br>$n = \frac{-3 \pm \sqrt{385}}{2}$<br>$n = -11.3107...$ or $n = 8.3107....$<br>Since $n$ is a positive whole number, **8 items**.<br>Check: $M(8) = 188 \leq 200$; $M(9) = 228 > 200$. | 4 | 1 for the capacity inequality or boundary equation.<br>1 for solving the quadratic.<br>1 for applying the positive whole-number domain.<br>1 for 8, justified by the interval or the neighbouring memory values. |
| (c) | A **translation** $\frac{3}{2}$ units left and $\frac{15}{2}$ units up. | 2 | 1 for each correctly directed component. |

| Graphs of Quadratic Functions continued | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 3.(a) |  | 2 | 1 for at least two correctly plotted points on $y = 1 - x$.<br>1 for a straight line through them across the grid. |
| (b) | $\mathbf{(0, 1)}$ **and** $\mathbf{(3, -2)}$. | 2 | 1 for each coordinate pair. Allow sensible graph-reading tolerance. |
| (c) | $x^2 - 4x + 1 = 1 - x$<br>$x^2 - 3x = 0$<br>$x(x - 3) = 0$, so $\boldsymbol{x = 0}$ **or** $\boldsymbol{x = 3}$. | 2 | 1 for equating the functions and rearranging.<br>1 for both roots. |

| Geometry • 38 marks | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 1.(a) | $OA = OB$ because both are radii.<br>$\angle OAB = \dfrac{180° - 110°}{2} = \mathbf{35°}$. | 2 | 1 for 35°.<br>1 for equal base angles in an isosceles triangle and the triangle angle sum. |
| (b) | $\angle OAT = \angle OBT = 90°$.<br>$\angle ATB = 360° - 110° - 90° - 90°$<br>$\boldsymbol{\angle ATB = 70°}$. | 3 | 1 for the radius–tangent right angles.<br>1 for using the angle sum of quadrilateral OATB.<br>1 for 70°. |
| (c) | C lies on the minor arc AB, so angle ACB subtends the major arc AB. The corresponding reflex central angle is $360° - 110° = 250°$.<br>$\angle ACB = \dfrac{1}{2}(250°) = \mathbf{125°}$. | 2 | 1 for 125°.<br>1 for the angle at the circumference being half the central angle on the same arc, correctly using the major arc. |
| 2.(a) | $\overrightarrow{AB} = \boldsymbol{q} - \boldsymbol{p}$<br>$\overrightarrow{OC} = \boldsymbol{p} + \dfrac{2}{3}(\boldsymbol{q} - \boldsymbol{p})$<br>$\boldsymbol{\overrightarrow{OC} = \frac{1}{3}p + \frac{2}{3}q}$. | 2 | 1 for the correct fraction of AB.<br>1 for OC. |
| (b) | $\overrightarrow{AC} = \dfrac{2}{3}(\boldsymbol{q} - \boldsymbol{p})$<br>$\overrightarrow{AD} = (2\boldsymbol{q} - \boldsymbol{p}) - \boldsymbol{p}$<br>$= 2(\boldsymbol{q} - \boldsymbol{p}) = 3\overrightarrow{AC}$.<br>The vectors share starting point A and are parallel, so A, C and D are collinear. | 2 | 1 for a correct comparison of AC and AD.<br>1 for the common-starting-point collinearity conclusion. |
| (c) | $\overrightarrow{CD} = \overrightarrow{AD} - \overrightarrow{AC} = 2\overrightarrow{AC}$<br>$\boldsymbol{AC : CD = 1 : 2}$. | 2 | 1 for CD being twice AC.<br>1 for ratio 1 : 2. |

| Geometry continued • additional vector questions | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 3.(a) | $\overrightarrow{OP} = \frac{2}{3}\boldsymbol{p}$<br>$\overrightarrow{OQ} = \frac{2}{3}\boldsymbol{q}$ | 2 | 1 for each vector. |
| (b) | $\overrightarrow{OR} = \frac{1}{2}\left(\overrightarrow{OA} + \overrightarrow{OQ}\right)$<br>$= \frac{1}{2}\boldsymbol{p} + \frac{1}{3}\boldsymbol{q}$<br>$\overrightarrow{OS} = \frac{1}{2}\left(\overrightarrow{OB} + \overrightarrow{OP}\right)$<br>$= \frac{1}{3}\boldsymbol{p} + \frac{1}{2}\boldsymbol{q}$ | 2 | 1 for each midpoint position vector. Accept equivalent unsimplified expressions. |
| (c) | $\overrightarrow{RS} = \overrightarrow{OS} - \overrightarrow{OR}$<br>$= \frac{1}{6}(\boldsymbol{q} - \boldsymbol{p})$<br>$\overrightarrow{AB} = \boldsymbol{q} - \boldsymbol{p}$<br>Thus $\overrightarrow{RS} = \frac{1}{6}\overrightarrow{AB}$.<br>$\boldsymbol{RS \parallel AB}$ **and** $\boldsymbol{RS : AB = 1 : 6}$. | 2 | 1 for the correct vector RS.<br>1 for comparison with AB, the parallel conclusion and ratio $1 : 6$. |
| 4.(a) | $\overrightarrow{PQ} = \frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$<br>$\overrightarrow{SR} = \frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$ | 2 | 1 for each vector. Direction SR, not RS, is required. |
| (b) | $\overrightarrow{PQ} = \overrightarrow{SR}$, so PQ and SR have equal length and are parallel in the same direction.<br>One pair of opposite sides of PQRS is equal and parallel.<br>**PQRS is a parallelogram**. | 2 | 1 for the directed-vector comparison.<br>1 for the correct parallelogram criterion and conclusion. |
| (c) | From A:<br>$\overrightarrow{AP} = \frac{1}{2}\boldsymbol{a}$, $\overrightarrow{AR} = \frac{1}{2}\boldsymbol{a} + \boldsymbol{b}$;<br>midpoint of PR has vector $\frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$.<br>$\overrightarrow{AQ} = \boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$, $\overrightarrow{AS} = \frac{1}{2}\boldsymbol{b}$;<br>midpoint of QS has vector $\frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$.<br>Since $\overrightarrow{AC} = \boldsymbol{a} + \boldsymbol{b}$, both are the midpoint of AC. | 2 | 1 for the midpoint of PR.<br>1 for the midpoint of QS and comparison with the midpoint of AC. A statement that diagonals bisect each other without the requested position-vector working is insufficient. |

| Geometry continued • additional vector questions | | | |
|---|---|---|---|
| **Question** | **Working and answer** | **Marks** | **Comments** |
| 5.(a) | $\overrightarrow{AP} = \boldsymbol{a} + \frac{2}{5}(\boldsymbol{b} - \boldsymbol{a})$<br>$= \frac{3}{5}\boldsymbol{a} + \frac{2}{5}\boldsymbol{b}$<br>$\overrightarrow{BQ} = -\boldsymbol{a} + \frac{4}{5}\boldsymbol{b}$ | 2 | 1 for each vector. |
| (b) | $\overrightarrow{AT} = k\left(\frac{3}{5}\boldsymbol{a} + \frac{2}{5}\boldsymbol{b}\right)$<br>$\overrightarrow{AT} = \boldsymbol{a} + m\left(-\boldsymbol{a} + \frac{4}{5}\boldsymbol{b}\right)$<br>Comparing coefficients:<br>$\boldsymbol{\frac{3}{5}k = 1 - m}$ and $\boldsymbol{\frac{2}{5}k = \frac{4}{5}m}$. | 2 | 1 for each correct coefficient equation, supported by the two vector expressions. Allow follow-through from (a). |
| (c) | $m = \frac{k}{2}$<br>$\frac{3}{5}k = 1 - \frac{k}{2}$, so $\frac{11}{10}k = 1$.<br>$k = \frac{10}{11}$, so $AT = \frac{10}{11}AP$ and $TP = \frac{1}{11}AP$.<br>$\boldsymbol{AT : TP = 10 : 1}$. | 2 | 1 for $k = \frac{10}{11}$.<br>1 for ratio 10 : 1, not 10 : 11. |
| 6.(a) | $\overrightarrow{OB} = 2\boldsymbol{a} + \boldsymbol{b}$<br>$\overrightarrow{AC} = -6\boldsymbol{a} + \boldsymbol{b}$ | 2 | 1 for each vector. |
| (b) | $\overrightarrow{OX} = s(2\boldsymbol{a} + \boldsymbol{b})$<br>$\overrightarrow{OX} = 6\boldsymbol{a} + t(-6\boldsymbol{a} + \boldsymbol{b})$<br>Coefficients of $\boldsymbol{a}$: $2s = 6 - 6t$.<br>Coefficients of $\boldsymbol{b}$: $s = t$.<br>$8s = 6$, hence $\boldsymbol{s = t = \frac{3}{4}}$. | 3 | 1 for each coefficient equation derived from the two vector routes.<br>1 for solving and stating both parameters. |
| (c) | $\overrightarrow{OX} = \frac{3}{2}\boldsymbol{a} + \frac{3}{4}\boldsymbol{b}$.<br>As $MN \parallel OA$, $\overrightarrow{OM} = \frac{3}{4}\boldsymbol{b}$.<br>Therefore $\boldsymbol{\overrightarrow{MX} = \frac{3}{2}a}$.<br>On AB, write $\overrightarrow{ON} = 6\boldsymbol{a} + u(-4\boldsymbol{a} + \boldsymbol{b})$.<br>Its $\boldsymbol{b}$ coefficient is $\frac{3}{4}$, so $u = \frac{3}{4}$ and $\overrightarrow{ON} = 3\boldsymbol{a} + \frac{3}{4}\boldsymbol{b}$.<br>Therefore $\boldsymbol{\overrightarrow{XN} = \frac{3}{2}a}$.<br>Since $\overrightarrow{MX} = \overrightarrow{XN}$, **X is the midpoint of MN**. | 2 | 1 for MX.<br>1 for XN with a valid midpoint conclusion. Accept an equivalent method using segment ratios. |

**Total: 160 marks**